

Gautham Kailash

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EDUCATION

National University of Singapore

Graduating May 2026

Bachelor of Computing in Computer Science, GPA: 4.62/5.00 (First Class Honours)

Singapore

- **Awards:** Recipient of the SoC Venture Initiation Program Grant, valued at \$10,000
- **Special Programs:** Student at NUSC, the honours college of NUS
- **Teaching Experience:** Teaching Assistant for CS2030S Programming Methodology II
- **Leadership:** VP of Pre-100 Club, Faculty Coordinator for NUSC Mentorship Programme, Co-Founder of VoilaAI
- **Courses:** Computer Networks, Operating Systems, Parallel and Concurrent Programming

EXPERIENCE

Goldman Sachs

May 2025 – Aug. 2025

Incoming Software Engineer Intern

Singapore

- Working with the Margin Technology team (post-trade transactions) within Global Banking and Markets.

AI Singapore

May 2024 – Aug. 2024

Software Engineer Intern

Singapore

- Engineered a high-performance MERN backend for a code assessment platform, supporting over 800 concurrent users.
- Optimised backend algorithms and database queries, achieving a 25% reduction in system load times.
- Implemented JSON/CSV export, file uploads, and email reporting, reducing data management time by 30%.

Monster Group

May 2023 – Aug. 2023

Software Engineer Intern

Jakarta, Indonesia

- Developed Triple One Global Indonesia's website using Next.js, Typescript, Prisma, PostgreSQL.
- Executed efficiency measures including reusable components, resulting in a 50% increase in development speed.
- Engineered a content management system enabling non-technical members to update the website.

NUS FinTech Lab

Aug. 2022 – Apr. 2023

Frontend Developer and Research Intern

Singapore

- Developed new NUS FinTech Lab website, enabling seamless navigation and a 40% reduction in page load times.

Netcon Technologies

Mar. 2019 – Apr. 2019

Cyber Security Intern

Chennai, India

- Demonstrated proficiency in XArp, Wireshark, and Metasploit to execute rigorous testing on company projects.

PROJECTS

KaithanOS | <https://github.com/kaithanos/kaithanos>

Jan. 2024 – Present

- Developed and implemented an OS bootloader in C, ensuring efficient initial hardware and system software loading.
- Integrated the FAT file system into the OS, enabling disk read/write capabilities and supporting FAT subdirectories.
- Engineered interrupt handling with IDT/GDT and optimised memory management for enhanced performance.

Concurrent Systems | <https://github.com/kailashgautham/concurrent-systems>

Jan. 2024 – May 2024

- **C++:** Developed a high-performance exchange matching engine, handling up to 100 threads and 4 million orders.
- **Go:** Built a concurrent exchange matching engine using channels and goroutines, achieving instrument-level concurrency.
- **Rust:** Engineered a multi-client TCP server with Tokio for async task management, achieving a 30x speedup.

InterviewAce | <https://github.com/janniver/interviewace>

Jan. 2024 – Present

- Developed an AI-powered platform with live coding, real-time speech, and video feedback to simulate interviews.
- Programmed in React, Python and Flask using Whisper and GPT-4 Turbo for advanced speech and language processing.
- Optimised platform responsiveness with advanced audio libraries and speech recording algorithms.

StealthSuite | <https://github.com/kailashgautham/stealthsuite>

Dec. 2023 – Present

- Developed StealthSuite, a comprehensive privacy application enabling users to anonymise images, documents, and audio, enhancing personal data protection across digital platforms (Express, React, Node, PostgreSQL)
- Implemented advanced features including face blurring, sensitive information redaction, and secure sharing.

TECHNICAL SKILLS

- **Languages:** C++, C#, Rust, Go, Java, Python, C, JavaScript, TypeScript, SQL
- **Frameworks/Platforms:** React, Node, Express, Nest, SciKitLearn, Unity
- **Developer Tools:** Git, Vim, Unix, Bash